Air Liquide
HEALTHCARE



Company Profile

www.in.airliquide.com

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ABOUT AIR LIQUIDE

Air Liquide is a world leader in gases, technologies and services for Industry and Health. Oxygen, nitrogen and hydrogen are essential small molecules for life, matter and energy. They embody Air Liquide's scientific territory and have been at the core of the company's activities since its creation in 1902.

Inventors of the future

Our ambition is to be a leader in our industry, to deliver long term performance and to contribute to a more sustainable world.

Through our positions in key markets of the future, we are at the heart of today's and tomorrow's challenges: energy and environmental transition, transformation of healthcare and technological progress. Our business expertise, our capacity for innovation, our proximity to our customers and patients, and the commitment and professionalism of our employees enable us to provide concrete solutions for the benefit of all our stakeholders and, beyond, of society.

A collective energy to serve our 3.8 million clients and patients

- 66,400 employees in 75 countries committed on a daily basis to better listen to and serve our customers and patients
- 2+ Million industrial customers all over the world Craftsmen, SMEs, ETIs, multinationals from sectors as varied as steel, energy, chemicals, car manufacturing, aeronautics, food processing or the pharmaceutical industry.
- 1.8+ Million patients that we support in hospitals and at home, alongside health professionals and structures

Air Liquide Healthcare

As the Healthcare activity of the Group, Air Liquide Healthcare is engaged alongside patients, healthcare professionals and hospitals to make the healthcare system efficient and virtuous for all. All this is already a work in progress, with one imperative: create value for the patient. With 15,600 men and women, we're a major world player in healthcare, an expert in chronic diseases follow up at home and new places of care, and supplier of services and medical gases solution for hospitals in particular.

In hospitals, whether it's in the emergency room, operating theater or intensive care unit, Air Liquide Healthcare medical gases solutions are valuable to help patients, often in critical moments. Our teams work alongside healthcare professionals to develop solutions that help them take care of their patients and offer associated services that enable them to relieve pain, anesthetize and improve respiratory functions.

Medical gases and associated services supplied by Air Liquide enable professionals in hospitals to care, relieve pain, anesthetize and improve respiratory functions. Medical gases in our plants are manufactured under strict process control ensuring traceability and quality in accordance with regulations.

Strategically Located Production Plants

Air Liquide started its operation in India in 1992. It is a predominant player in the Indian Market with state-of-the-art infrastructure in gas production and distribution. In North India, Air Liquide caters to its industrial customers and hospitals through its plants located in Panipat, Roorkee & upcoming new site in Kosi near Mathura. With a total installed capacity of more than 1000 Tpd and Liquid storage capacity of more than 5000 Tonnes, Air Liquide is the largest gas manufacturer in North India. The new plant in Kosi, Mathura is expected to be operational before the end of 2023.

Air Liquide is also an established player in West India with its Air Separation Units in Jhagadia, near Bharuch and in Chakan Pune. In Dec 2022 with the startup of Nagpur plant, Air Liquide commenced operations of its 3rd manufacturing unit in West.

Air Liquide with its fleet of 100+ tankers and Semi Trailers, is considered the most reliable supplier delivering products across India.

Air Liquide contribution during Covid-19

Being the largest manufacturer of Oxygen in North India, Air Liquide had shoulder a big responsibility during the 2nd phase of Covid-19, when the nation was struggling due to an exponential increase in the demand of Oxygen. Air Liquide ensured that all the volumes allocated by the GoI were strictly adhered to.

A snapshot from Times of India depicting our capability to ensure an uninterrupted supply of Medical Oxygen even during the second wave of Covid-19 pandemic.

OXYGEN RECEIVED IN DELHI TILL MIDNIGHT FROM APRIL 28 TO MAY 4

Name Of Manufacturer	Location	Allocated	April 28	April 29	April 30	May 1	May 2	May 3	May 4
			Received	Received	Received	Received	Received	Received	Received
Air Liquid	Panipat	170	187	189.2	101.7	194.7	150	154.7	176.9
	Roorkee	20	0	0	0	0	17.5	12	0
Linde	Jamshedpur-30 Rourkela-40 Selaqui (Faridabad)	255	79.2	98.6	118.3	95.1	90.7	75.9	200.2
	Durgapur-30		0	0	0	0	90	0	0
Inox	Surajpur	35	21.9	23.2	18.8	24.7	28	40.6	30.3
	Modi Nagar	30	71.5	47.5	43.3	65.1	62	63.5	62.1
	Bartiwala	20	0	0	0	0	0	0	0
Goyal Glasses	Ghaziabad	30	32	24.8	42.4	21.9	9.2	6.2	18.7
India Glycols	Kashipur	30	39.5	26	0	20.3	0	0	16.5
Jindal Steel Power	Raipur		0	0	0	0	0	0	0
One time additional supply of 80MT from GOI			0	0	0	0	0	80	0
Additional Supply Of 80 MT from Reliance, Jamnagar			0	0	0	0	0	0	80
Seth Traders Through Linde	Faridabad		0	0	0	0	0	0	0
	Total	590	431.2	409.3	324.6	422	447.5	433	585

Data by central government; Prior allocation to Delhi as of May 1: 490 MT;
Note1: Additional supply of 80MT From Reliance, Jamnagar through train; **Note2:** Supply of 120MT from M/S Linde through train

The challenges you face every day related to gas management are well understood including, your efforts to be,

- 100% ready to address any emergency situation, which can impact **patient safety**
- sure that enough **oxygen pressure** is maintained across the MGPS network
- able to **switch sources** in case of drop in pressure
- sure of having adequate stocks to **avoid dry-outs** incase **demand increases** or until next supply
- confident that there is zero probability of **supply interruptions**

All of these routine tasks prevent you from focusing on core activities, while adding substantial stress and unnecessary costs in managing the associated risks.

In this proposal, you'll find more information about your Medical Oxygen Supplies, a unique offer **designed in accordance with ISO 7396-1 and HTM 02-01 for Continuity of Supplies** in order to ensure highest level of Patient Safety and Optimization of your gas supply costs. This offer is,

☑ Designed to ensure Continuity of Supplies in normal and single fault conditions (LLG Configuration)
☑ Tank Sizing in accordance with HTM 02-01 aligned with distance from nearest source
☑ Groups of alarms in line with ISO 7396-1
☑ Automatic switchover between sources as per ISO 7396-1 recommendations
☑ Allows you to follow the evolution of your gas consumption any time through the MyGas digital portal
☑ Value added services including Annual Demand validation, Supply Interruption process, Mock Drills

Based on findings from the Preliminary Risk Identification survey done at your site, and demand profile shared by you, Air Liquide is pleased to submit this Proposal to help you meet your intended KPIs.

Implementing the Air Liquide Offer will not only improve reliability of gas supplies exponentially, but will **reduce the safety risks, costs and time associated with maintaining the secondary backup system, which is included in our offer.**

In the event of normal or single fault conditions, if supplies from Primary source are interrupted, to ensure continuity of supplies, as per ISO 7396-1 **Secondary backup Source** should **automatically** commence the supplies. The Secondary Source should also be permanently connected to the MGPS, and should be able to supply the **whole distribution system**.

Activation of the **reserve supply** is expected , in case both the Primary and the Secondary Sources are exhausted or during maintenance. This source switchover may be **automatic or manual.**

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PROPOSITION

Consider implementing the New Air Liquide Liquid Medical Oxygen Solution, a unique offer designed in accordance with ISO 7396-1 and HTM 02-01 for Continuity of Supplies in order to ensure highest level of **Patient Safety and optimization of your gas supply costs.**

The offer includes **LLG configuration**, where Primary & Secondary sources are Liquid and these sources shall be owned, installed & maintained by Air Liquide. The Reserve Source may be in Cylinders and is excluded from this offer, and shall be in Hospital's scope.



Continuity of Supplies
Both Primary & Secondary sources with automatic switch over

Low Operational Expenses
Secondary Cylinder Backup not required; Minimize Cylinder Handling Risks

Enhanced Storage Capacity
~9 days product in primary & ~2 days stock in secondary tank, Lesser deliveries reduces associated safety risks

Value Added Services
Annual demand validation, Mock drills with Hospitals, Supply Interruption.

Alarms to Hospital
Protection as per ISO 7396 -1, Audio & Visual Alarms available to hospital

Digital Interface
Access to MyGas Portal for all gas related queries and supply informations

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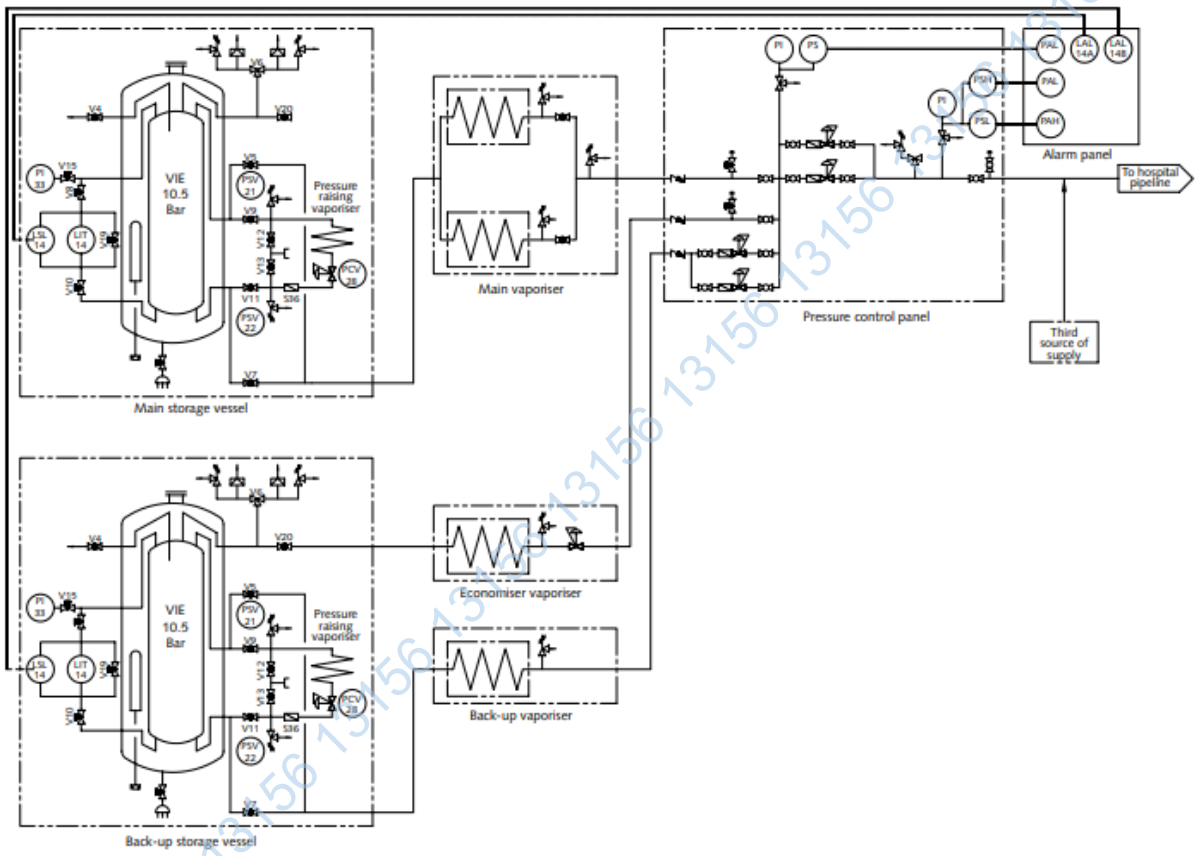
Details of Air Liquide offer

The LLG configuration for the medical Oxygen supplies is the most reliable and economical solution for **all medium and large hospitals** having 200 beds or more. The primary & secondary sources are supplied using Liquid Medical Oxygen, permanently connected to the main network, and designed for automatic supply to the complete pipeline network.

The primary and secondary liquid source shall be installed and maintained by Air Liquide. Reserve source and is not included in our offer. It may cover the complete pipeline system, or be connected to individual manifolds on each zone (or building) in order to provide additional protection against a pipeline failure within the facility.

As per ISO 7396-1 the proposed installation shall be equipped to detect (local alarms) and report to a 24/7 monitored system (e.g. control room, service center) incase one of the sources is not operational (e.g. low level, low pressure or out of service) or the system switched from the main source to the second one or in case the facility's supply pressure is above or below 20 % of the normal value.

Figure 19 Primary and secondary VIE system on single plinth



(Source : **Health Technical Memorandum 02-01 : Medical Gas Pipeline system - Part A**)

Disclaimer *: Above typical schematic provides general recommendations by HTM 02-01 standards. Actual scope offered by Air Liquide shall differ based on the scope agreed mutually between the hospital and based on experience of Air Liquide in designing & managing such solutions.*

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Air Liquide shall also install a telemonitoring system to assess the stock levels of each tank and detect the switchover from primary to secondary source. The offer also includes access to MyGas Customer Portal, where all information regarding stocks, invoices can be accessed.

In addition to the **Financial Benefits,** the main Technical features and benefits of the solutions offered by Air Liquide are grouped under 3Ps

Patient Safety, Process Safety & Product Reliability

❑ Patient Safety

Description of the solutions	Benefits
❑ System designed with **three (3) independent Supply Sources**. **Primary & Secondary Source included** in the offer. Reserve Source by the Hospital	☑ Continuity of supplies assured with dual redundant, Fail Safe design ☑ Oversized Vaporisers to handle intermittent surge in oxygen demand. ☑ Consistent Product Quality assured
❑ **Highest Tank Autonomy levels** offered in the industry to avoid frequent refilling and associated risks during transit and tank filling (maximum 3 - 5 deliveries per month) ~ **9 days stock** considering Operation Stock, Safety Stock & Unusable stock levels ~ **2 days stock** in Secondary tank to allow adequate time to plan and react incase of single fault in the Primary Source	☑ Main supplies from Primary tank having 9 to 12 days of stock, depending on distance from nearest AL source ☑ Safety stock of 3 days always maintained in Primary Tank; Any breach in this level treated and reported as safety incident for investigation. ☑ Mandatory 2 to 3 days stock in Secondary backup tank for automatic supply due to any fault in the Primary supply line

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❑ Process Safety

Description of the solutions	Benefits
❑ **100% stock** always in Secondary Tank which shall be used only when Primary safety stock is exhausted Liquid vaporized in Secondary source fed to the main pipeline system to avoid wastage	☑ Continuity of supply assured in **normal and single fault** condition during maintenance ☑ **Dual Redundant** supply sources with independent primary & secondary storage tank circuit in accordance with ISO 7396-1
❑ Continuous supply assured with **Automatic Switchover** between different sources when pressure drops *(Reserve source not included in Air Liquide scope)*	☑ When line pressure drops below set point supply shift to automatically to Secondary tank through a pressure regulating device ☑ If line pressure further drops reserve source to come online
❑ Protection against potential with **Audio Visual Alarms** to Hospital control room 7 Nos. Audio/visual alarms for all emergencies envisaged in ISO 7396-1	☑ List of Alarms 1. Primary tank pressure is lower than normal 2. Primary tank level reach Safety Stock level 3. Primary tank is empty, secondary tank is in operation 4. Secondary tank level reach Safety Stock level 5. Secondary tank is empty & Reserve tank in operation 6. Pipeline pressure drops below 20% of normal pressure 7. Pipeline pressure increases above 20% of normal pressure
❑ **Installation Safety is predominant** and Air Liquide installations. All installations assessed for redundancy of **Elements important** for **Safety** (EIS tags)for safety of Hospital /Air Liquide personnel at site.	☑ **Backflow prevention** in the event of a single fault in either system ☑ **Thermal relief valves** to vent off liquid trapped between two valves, included which is not offered by any other player in the Industry ☑ **Lock Out Tag Out (LOTO)** mechanism followed for all critical equipments in the installation for Continuity of Supplies ☑ **EIS tagging** done to identify risks based on probability & severity of occurrence and mitigate the risks to reach acceptable levels

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❑ Product Reliability

Description of the solutions	Benefits
❑ **Traceability** of Product	☑ Oxygen IP (Medicinal Grade In Liquid Form as per Indian Pharmacopeia) with Purity of >99% ☑ Certificate of Conformance for agreed quality of product (purity level, moisture / impurity etc)
❑ Superior **Customer Experience**	☑ Regular monitoring to identify demand variation and develop mitigation strategy along with Hospital to ensure patient Safety ☑ Mock Drills with customer team to check the installation integrity during emergencies☑
❑ Digital interface using **MyGas Portal** provide 24/7 information about medical oxygen to hospitals **Telemonitoring** stock level	☑ Storage tank Levels for both Primary & Secondary tanks updated on daily basis ☑ Minimize order generation, processing and tracking ☑ More time to focus on your core activities
❑ Safety & operational training for **Hospital maintenance team** **Trained Air Liquide team** on criticality of medical installations to manage supplies	☑ Joint risk assessment prior to any scheduled supply interruption ☑ All Hospitals classified as Critical and prioritized. Drivers & Schedulers specially trained in managing hospital supplies. ☑ Yearly preventive maintenance of the installation to ensure 100% reliability

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